

Enhancing Usable Security and Privacy of Cryptocurrencies in the Quantum Age

1. INTRODUCTION

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ABSTRACT

Cryptocurrencies like Bitcoin and Ethereum are poised to form a foundation of future finance, but their mainstream adoption is impeded by critical usability and security challenges. Users today struggle with managing cryptographic keys and maintaining privacy, leading to frequent losses and breaches. The complex nature of cryptocurrencies and the unfamiliarity of decentralized blockchain finance makes consumers weary of financial use. At the same time, rapid advances in quantum computing threaten to undermine the cryptographic standards protecting these systems (e.g., ECDSA, RSA). This paper explores how to enhance the usable security and privacy of cryptocurrencies in light of the looming quantum era. We review the state of usable security in cryptocurrencies, highlighting issues such as poor wallet UX, human errors in key management, and privacy misconceptions. We then assess the current progress in quantum computing and its realistic threats to blockchain security, detailing how Shor's and Grover's algorithms could break popular digital signature schemes and weaken hash and symmetric algorithms. Quantum-vulnerable methods (ECDSA, RSA, etc.) are discussed alongside emerging post-quantum cryptography (PQC) solutions. We survey ongoing developments in PQC – including lattice-based and hash-based signatures – that are practical for blockchain use, and review transition strategies for upgrading existing cryptocurrencies to quantum-resistant schemes. Throughout, we emphasize human-centric security design, examining how wallet design, key recovery mechanisms, and user education can be improved to facilitate security without sacrificing usability. We present data-driven insights and include figures illustrating the scope of the problem and the effectiveness of proposed solutions. Our analysis, drawing on insights from 22 recent studies, underscores that proactively integrating usable security frameworks with post-quantum solutions is essential to safeguard the future of decentralized finance. We conclude with recommendations for researchers and practitioners to collaboratively advance the security, privacy, and usability of cryptocurrency systems before large-scale quantum computers become a reality.

Cryptocurrencies like Bitcoin and Ethereum have introduced a new way to exchange value—peer-to-peer and without relying on banks or other central authorities. Since Bitcoin's launch in 2009, these systems have evolved into a key pillar of decentralized finance, gaining traction among both individuals and institutions. By 2016, Bitcoin already had a market cap exceeding \$3.5 billion and supported over 130,000 transactions daily [1]. Since then, the crypto market has exploded, with thousands of coins in circulation and blockchain-based technologies influencing industries from finance to supply chain.

Despite this momentum, cryptocurrencies still haven't gone fully mainstream—especially for everyday transactions. A major reason is the steep learning curve and security risks associated with their use [2]. Unlike traditional banking systems, where institutions handle things like account recovery and fraud protection, cryptocurrency shifts those responsibilities directly onto the user. Managing a wallet, safeguarding private keys, and understanding privacy protocols are all essential—but many users find them confusing or intimidating [3][4].

2. BACKGROUND

3. PRIVACY ISSUES

Privacy is another concern closely tied to usability and user education. Cryptocurrency transactions are recorded on public ledgers, yet many users mistakenly believe Bitcoin is “anonymous”. In reality, Bitcoin offers pseudonymity at best; reusing addresses or failing to use mixing techniques can easily link transactions to an identity. The aforementioned study noted *“significant misconceptions on how to remain anonymous and protect privacy”* among users. A lack of easy-to-use privacy tools (or poor understanding of them) means that typical users often **fail to protect their financial privacy**. Dedicated “privacy coins” like Monero or Zcash have been developed to address these concerns, but they introduce additional complexity (e.g. managing view keys or shielded addresses) that can deter casual users. Thus, a challenge remains to **provide strong privacy guarantees in a user-friendly manner** within mainstream cryptocurrencies.

4. SECURITY OF CURRENT CRYPTOGRAPHY

Compounding these usability-driven issues is a looming technological threat: **quantum computing**. Blockchain security today relies heavily on cryptographic algorithms like the Elliptic Curve Digital Signature Algorithm (ECDSA) for transaction authentication and RSA/ECDH for some layer-2 protocols. The hardness of these algorithms (discrete logarithm and integer factorization problems) underpins the assumption that only someone with the private key can authorize a transaction. However, with the advent of quantum computing, this assumption no longer holds indefinitely. Quantum algorithms—most famously Shor’s algorithm—can theoretically solve these problems **in polynomial time**, rendering **RSA, ECDSA, and other public-key algorithms breakable** if a sufficiently powerful quantum computer is built. Likewise, Grover’s algorithm can speed up brute-force search, undermining symmetric ciphers and hash functions by effectively halving their security strength. In essence, the cryptographic foundation of current cryptocurrencies is **quantum-vulnerable**.

Emerging Quantum Threat: While large-scale, error-corrected quantum computers do not yet exist, progress in quantum computing has been rapid. In 2019, Google’s 53-qubit Sycamore processor achieved a milestone known as “quantum supremacy,” performing a special-purpose computation infeasible for classical supercomputers. Since then, quantum hardware has advanced steadily. IBM unveiled a 127-qubit processor in 2021 and a 433-qubit processor in 2022, representing a tripling of qubit count in one year. They aim to reach over a thousand qubits in the next generation. Although these machines are still far from the many thousands of *error-corrected* qubits required to threaten 2048-bit RSA or 256-bit elliptic curves, the trajectory is clear. As a 2022 scientific report cautioned, “*current quantum computers have already proven themselves able to break short asymmetric keys... it is only a matter of time before robust quantum computers... will be able to break larger and larger keys*”. Government agencies and cryptographers take this threat seriously: the U.S. National Institute of Standards and Technology (NIST) has initiated a program to standardize **post-quantum cryptography (PQC)** by 2024-2025, and suggests deprecating traditional crypto by 2030. The U.S. National Security Agency (NSA) likewise recommends transitioning to quantum-resistant algorithms in the next few years and using them exclusively by the early 2030s. This looming deadline adds urgency to the issue – a concept often summarized as “**harvest now, decrypt later**”, meaning an adversary could record encrypted or signed data today and decrypt it in the future once quantum capabilities are available.

In summary, cryptocurrency adoption faces a dual challenge: improving **usable security and privacy** for users in the present, while **future-proofing** systems against quantum threats tomorrow. These challenges are interrelated; for example, any transition to post-quantum cryptography must keep usability in mind so that users will actually adopt it. This paper presents a comprehensive review and analysis of approaches to tackle

both facets. We first examine the current state of security and privacy in cryptocurrencies from a usability perspective, drawing on empirical research and incident reports. Then, we assess the state of quantum computing and detail the vulnerabilities in blockchain protocols (and users’ assets) that quantum attacks could exploit. Finally, we explore solutions: from near-term improvements in wallet design and user education, to long-term migration to post-quantum cryptographic algorithms and protocols. We highlight ongoing developments in quantum-resistant blockchains and the importance of integrating these advances in a user-centric way. Our goal is to illuminate a path by which cryptocurrencies can remain secure and private **by design** – not only against today’s threats, but also against the next generation of adversaries – all while being **usable** enough to facilitate mass adoption.

5. RELATED WORK

A number of existing works have addressed aspects of cryptocurrency security, privacy, and the quantum threat, though often separately. In terms of **cryptocurrency security and privacy**, there have been several surveys and studies. Asadi *et al.* provide a comprehensive survey focusing on the security of Bitcoin and other cryptocurrencies, cataloguing known attack vectors such as 51% attacks, transaction malleability, and networking vulnerabilities. They emphasize that security and privacy remain major challenges for real-world cryptocurrency usage, and discuss countermeasures in the literature. However, they note a lack of surveys specifically centered on these security and privacy concerns. Our work helps fill that gap by focusing on how these issues intersect with usability and upcoming quantum risks.

On the usability side, Krombholz *et al.* conducted one of the first large-scale studies on Bitcoin user security and privacy experiences. They quantified user behaviors (as mentioned in the Introduction) and qualitatively explained why users make certain security decisions. Their findings (high usage of online wallets, under-utilization of security features, privacy misconceptions, etc.) highlight the usability shortcomings of current wallet software. Other researchers have proposed improvements to the user experience of cryptocurrency transactions. Kattwinkel and Rademacher (2020) analyzed common failure modes in personal Bitcoin payments and proposed a new payment protocol that exchanges preparatory information between sender and receiver to **reduce user errors and attack surface**, yielding a more **user-friendly yet secure** transaction flow. Their evaluation demonstrated significantly improved usability and security balance using this protocol. Such human-centric approaches inform our discussion on improving usable security.

In parallel, a growing body of work addresses the looming **quantum threat to cryptocurrencies**. Stewart *et al.* (2018) examined the threats a “quantum-capable adversary” would

pose to Bitcoin’s security model and proposed a protocol for migrating funds to quantum-safe addresses . Notably, their *commit–delay–reveal* scheme would allow users to move bitcoins from vulnerable ECDSA-based outputs to post-quantum secure outputs even **after** a quantum attack has begun, albeit slowly . This scheme, which could be deployed as a soft fork, is an important reference point for designing transition plans that are robust even under worst-case timing (we discuss this further in Section IV). Other researchers have taken a more proactive approach. **Post-quantum cryptocurrency schemes** have been proposed that build quantum resistance in from the start. For example, Gao *et al.* (2018) present a lattice-based signature scheme integrated into a blockchain, defining the notion of a *post-quantum blockchain (PQB)* . Their scheme uses short integer solution (SIS) lattice assumptions and introduces a double-signature mechanism to reduce linkability between messages and signatures. They show that using this post-quantum cryptography, the blockchain can resist quantum attacks with acceptable performance trade-offs . Similarly, in 2023 Allende *et al.* demonstrated an **end-to-end quantum-resistant blockchain framework** on an Ethereum-like network . They combined multiple layers of defense: quantum random number generation for keys, post-quantum TLS for node communication, Falcon-512 post-quantum digital signatures for transactions, and on-chain smart-contract verification of those signatures . This practical implementation showed that retrofitting an existing blockchain with post-quantum security is feasible . We build upon these works by examining how such solutions can be integrated while preserving or even enhancing usability.

There are also conceptual works exploring **fully quantum or hybrid quantum blockchain systems**. Jogenfors (2016) introduced “Quantum Bitcoin,” a theoretical currency secured by the no-cloning theorem of quantum mechanics rather than traditional cryptography . In that scheme, quantum states (quantum information) would represent coins, and double-spending is prevented by the physics of quantum observations . While intriguing, such approaches require quantum technologies that are not yet mature (like quantum memory and communication networks for end-users). They also raise new usability questions (how does a user store a quantum coin?). Nonetheless, these ideas underscore that quantum technology can be **not only a threat but also a tool** for security . Our work remains grounded in the assumption that classical blockchains will need to adopt post-quantum *classical* cryptography, rather than rely on full-blown quantum cryptography, as the latter is far from practical for broad deployment . NSA, for instance, views post-quantum cryptography as a more practical defense than quantum key distribution for most applications .

Finally, in the broader **quantum cybersecurity** context, surveys by Faruk *et al.* (2022) and Singh & Kumar (2024) provide overviews of how quantum computing impacts cybersecurity and what countermeasures are emerging . They reinforce the notion that quantum computing poses “*the most unexpected threats*” to current cryptosystems, but also highlight “*quantum-resistant encryption algorithms*” and even quantum machine learning as potential solutions . These works, however,

do not delve deeply into cryptocurrencies specifically. By synthesizing insights from all these areas – usable security, blockchain attacks, quantum threats, and post-quantum solutions – our paper aims to provide a unified analysis relevant to securing cryptocurrencies in the quantum age.

6. USABLE SECURITY AND PRIVACY IN CRYPTOCURRENCY

Cryptocurrencies flip the traditional security model of finance on its head: **users hold the keys** (literally) to their funds and are responsible for their own security. This empowerment comes at a cost. As noted, many users are ill-equipped to manage high-stakes cryptographic secrets. In this section, we examine common usability pain points and their security implications, as well as ongoing efforts to address them.

Key Management and Wallet Usability

Managing private keys is arguably the Achilles’ heel of cryptocurrency usability. A string of 64 hexadecimal characters (a typical representation of a 256-bit private key) is not something an average person can handle reliably. Wallet software attempts to abstract keys into seed phrases, password-encrypted files, or hardware devices, but each approach has usability trade-offs. Users must safely **back up their keys/seed** and prevent loss or theft. Failure to do so means irreversible loss of funds – there is no “password reset” in Bitcoin. This is a stark departure from traditional banking and places a cognitive load on users. Indeed, *reports from online forums abound of users irretrievably losing coins due to lost keys or failing to back up wallet files* . Human error is a leading cause of cryptocurrency loss.

Several strategies have emerged to ease key management. **Hierarchical Deterministic (HD) wallets** use a single master seed to generate an infinite sequence of addresses, so that one backup (a 12- or 24-word mnemonic) can secure an entire wallet. This is a significant usability improvement—users only need to record one seed phrase. However, writing down a recovery phrase and keeping it safe is still not foolproof. Users may not understand it, may lose the paper, or have it stolen if not concealed. Some wallets now offer **cloud-based key backups** or social recovery (Shamir-secret-sharing the seed among friends), but these introduce new trust and security considerations. Krombholz *et al.* observed that even with HD wallets, many users “*do not use all security capabilities of their selected tool*” —for example, not enabling passphrase encryption on the seed, or not using the option to make paper backups at all. In their interviews, users expressed that certain wallet procedures (like manually writing down seeds or using “paper wallets”) felt cumbersome or strange . There is clearly a need to design backup and recovery mechanisms that align better with user behavior and mental models.

Another usability vs. security trade-off is the choice between **software wallets, hardware wallets, and custodial solutions**. Hardware wallets (secure devices that store keys offline) are

considered a gold standard for security, as they isolate private keys from potentially malware-infected computers. However, they add steps to the user experience (connecting a device, verifying addresses on a tiny screen) and cost money, which can deter casual users. Many users, as the studies show, default to the more convenient option: leaving coins on an exchange or online wallet. The result is effectively recentralization and single points of failure – *e.g.*, the infamous Mt. Gox exchange hack in 2014 led to ~\$470 million worth of Bitcoin being stolen from hundreds of thousands of users who had entrusted their keys to that platform. Despite such incidents, the allure of easy access and an interface with password recovery keeps drawing users to custodial services. The challenge ahead is to give users *Mt.Gox-level convenience without Mt.Gox-level risk*. This could involve wider use of hardware wallets (driving costs down and simplifying UX), or innovative approaches like **threshold signatures** (where keys are split so that even if a user loses one piece, an attacker cannot spend funds without the others). Some exchanges now offer multisig escrow where the user holds one key and the exchange another, so that theft of the exchange alone can't move funds – but again, this requires users to manage a key.

Transaction Management

Even when keys are safely stored, users face pitfalls in **constructing and handling transactions**. Sending cryptocurrency requires extreme precision – a wrong address or wrong network can lead to irrevocable loss. Human-readable addresses or domain-based identifiers (via ENS, FIO Protocol, etc.) are emerging to reduce errors, but many wallets still show raw addresses by default. Kattwinkel and Rademacher identify the step of **address exchange** between sender and recipient as a risky point, vulnerable to man-in-the-middle attacks (clipboard malware swapping out addresses, for instance) or user mistakes. Their proposed protocol has senders and recipients share payment information out-of-band in a secure way, effectively adding a verification step to ensure the address and amount are correct. Such workflows, if made seamless, can prevent common errors like sending to the wrong address or sending the wrong amount. Another issue is **fee management** – users often don't understand network fees and may underpay (leading to stuck transactions) or overpay. Modern wallets are improving by auto-estimating fees, but when the network is congested, this can become complex. Poor handling of fees and transaction confirmation feedback can confuse users and lead them to resend transactions, causing duplicates or simply anxiety.

Privacy and Anonymity Tools

On the privacy front, there's a usability challenge in empowering users with anonymity tools. Techniques like CoinJoin (mixing transactions among multiple users) or using privacy-enhanced cryptocurrencies require extra steps and knowledge. Krombholz *et al.* found widespread misconceptions about what it takes to be anonymous on Bitcoin – for example, some users thought simply not associating their name was enough, unaware of blockchain analysis techniques. This indicates a need for better **privacy-by-default** features. Newer wallets have started integrating CoinJoins or automated address rotation to help

privacy, but they often come at performance or convenience costs (CoinJoins can be slow and require other participants). *Usable privacy* in cryptocurrencies remains an open problem: how to allow average users to attain strong privacy without having to become experts in operational security. Some projects experiment with **zero-knowledge proof-based mixers** and one-click shielding of funds (as in Zcash shielded addresses), but adoption is limited when the UX is not streamlined.

It's also worth noting that **perceptions of security vs. usability** can differ among user groups. In interviews, Krombholz *et al.* noted that “*many participants stated that for fast proliferation of Bitcoin, simple and intuitive UIs are more important than security*”, drawing analogies to how personal computers spread despite users not understanding them fully. This mindset – prioritizing ease-of-use even at some expense of security – is prevalent. Users are willing to take risks (or are unaware of them) if the alternative is an interface they find too complex. Therefore, **security by design** must accommodate user behavior. For example, if users prefer web wallets for convenience, perhaps the solution is to **harden web wallets** (through sandboxing, multi-factor authentication, or client-side private key encryption) rather than simply insisting users run local full nodes, which many will not do. Researchers have suggested dedicated hardware or apps for key management with intuitive UI as a way forward – essentially creating appliances for cryptocurrency use that abstract cryptography away from the user. Such devices, combined with backup solutions (maybe integrating with secure cloud or physical vault services), could provide both usability and security – though cost and trust are factors to consider.

Summary of Usability Issues

In summary, the main usable security issues in cryptocurrencies include: (1) **Private key and seed management** – users risk loss/theft without user-friendly backup and storage methods, (2) **Wallet and user interface design** – complex interfaces and jargon lead to misuse or avoidance of security features, (3) **Transaction handling** – the lack of human-readable identifiers and confusion around fees/confirmations cause user errors, and (4) **Privacy** – achieving anonymity requires steps users often find too complex. Each of these has seen active research and development of mitigations, from HD wallets and hardware keys to better GUI design and mixing services. But bridging the gap between expert recommendations and actual user practices remains challenging. As one study succinctly put it, “*no amount of security will be enough*” if the private key is exposed – and ensuring that users can keep their keys confidential (and recoverable) is as much a usability problem as it is a technical one.

The Quantum Computing Threat to Cryptocurrencies

While the usability issues discussed above are pressing, they occur against a backdrop of cryptographic techniques that, for now, largely **do their job** when used correctly: ECDSA provides unforgeable signatures, hash functions provide immutability, and so on. However, the rise of quantum computing threatens to **fundamentally break the security assumptions** of current cryptocurrencies in the not-too-distant future. In this section, we outline the state of quantum computing, the specific threats it poses to blockchain systems, and the timeline and likelihood of those threats materializing.

Quantum Computing 101

Quantum computers leverage quantum-mechanical phenomena (superposition and entanglement) to process information in ways

classical computers cannot. A quantum bit or *qubit* can represent 0 and 1 simultaneously (superposition), and multiple qubits can exhibit entanglement where their states are correlated in ways impossible classically. By manipulating qubits through quantum gates, certain computations can be done exponentially faster than on conventional silicon. The most relevant algorithms here are **Shor's algorithm** (for integer factorization and discrete logarithms) and **Grover's algorithm** (for unstructured search). Shor's algorithm, discovered in 1994, showed that a quantum computer with enough qubits could factor large numbers and compute discrete logs in polynomial time – tasks that form the basis of RSA and ECC security. Grover's algorithm, on the other hand, provides a quadratic speedup for brute-force search, meaning it effectively cuts the exponent of symmetric ciphers (e.g., turning a brute force on a 128-bit key from 2^{128} steps to roughly 2^{64} steps).

For years, these algorithms were mostly of theoretical concern because quantum hardware was extremely limited (just a few qubits with high error rates). However, the hardware landscape has been changing. Figure 2 illustrates the **growth of quantum processor sizes** over recent years. From just 5 qubits in 2016 (IBM's first public quantum processor) to **127 qubits by 2021 and 433 qubits by 2022 (IBM's "Eagle" and "Osprey" chips)**, the scale has accelerated. While raw qubit count isn't the only factor (quantum error rates and connectivity matter too), it is a rough indicator of progress. If current trends continue, we may see quantum processors with thousands of physical qubits within this decade. Experts estimate that to break RSA-2048 or Bitcoin's secp256k1 ECDSA, a quantum computer might need on the order of several thousand **logical (error-corrected) qubits**, which could correspond to millions of physical qubits with current error correction overheads. This is still a significant leap from where we are, but not inconceivable in a multi-decade horizon. In fact, government agencies are planning as if such quantum capabilities will be operational by the 2030s.

Vulnerable Cryptographic Primitives

Nearly all cryptocurrencies rely on a small set of cryptographic primitives: hash functions (SHA-256 in Bitcoin, KECCAK-256 in Ethereum), symmetric key ciphers (used in wallet seed generation, but not heavily in protocols), and public-key algorithms (ECDSA for signatures, ECDH for key exchange in payment channels, etc.). Each of these faces different threats from quantum algorithms:

- **Digital Signatures (ECDSA, RSA, EdDSA):** These are vulnerable to **Shor's algorithm**. Bitcoin's security, for example, rests on the assumption that given a public ECDSA key (which is a point on the secp256k1 elliptic curve), an attacker cannot derive the private key (the discrete logarithm problem). A large quantum computer running Shor's algorithm would completely undermine this – it could derive private keys from public keys efficiently. Notably, in Bitcoin and many cryptocurrencies, an address is often the hash of a public key (to reduce risk exposure), but once a transaction is made from that address, the public key is revealed on-chain. This means *any* unspent output whose public key has been revealed (i.e., any reused address or spent output) would be vulnerable to theft if an attacker has a quantum computer that can perform ECDLP on secp256k1. The attacker could simply derive the private key and spend the funds. This is the nightmare scenario: **quantum-enabled coin theft**. RSA-based systems (less common in crypto, but used in some altcoins or older systems) have a similar fate. Even newer schemes like Ed25519 (used in some cryptos for

signatures) are based on elliptic curves and would fall to Shor's algorithm as well.

- **Hash Functions (SHA-2, SHA-3 family):** Quantum attacks on hash functions come mainly from Grover's algorithm, which can find a preimage in $\sqrt{N}$ steps instead of N (meaning it reduces the security by half the number of bits). For Bitcoin's SHA-256, which has a 256-bit output, Grover's algorithm would effectively reduce the security to 128-bit. In practice, 128-bit security is still considered strong – comparable to AES-128. So SHA-256 and similar hashes are **not completely broken** by quantum, but their margin of safety is reduced. More concerning is that Bitcoin's proof-of-work mining is essentially a brute-force hash inversion process. Grover's algorithm theoretically gives a quadratic advantage to a miner with a quantum computer, allowing it to search the SHA-256 space faster. In fact, it's noted that Grover could provide a *"quadratic advantage for mining blocks in certain consensus protocols such as proof of work"*. However, the practicality is debatable: one would need an extraordinarily fast quantum computer to outrun the combined hash power of classical miners, and Grover's advantage is only quadratic, which might not be decisive in the near term. Nonetheless, it's a vector to monitor – in a future where few entities have quantum machines, they could potentially dominate PoW mining. Hash-based Proof-of-Stake or other consensus mechanisms might thus become more attractive long-term.
- **Symmetric Ciphers (AES, etc.):** Similar to hashes, Grover's algorithm can brute force symmetric keys in $\sqrt{N}$ time. Most cryptocurrencies don't rely heavily on symmetric encryption at the protocol level (aside from things like LN channel secrecy or P2P message encryption). Wallets using AES-256 for encrypting private keys, for instance, would see their effective security drop to 128-bit, which is still robust. The bigger concern is more about **long-term encrypted storage** of crypto assets. If someone has an encrypted wallet backup (protected by, say, AES-128), a quantum attacker in the future could potentially decrypt it faster. But best practices already suggest AES-256, which even under Grover remains ~128-bit secure.

In summary, the **most urgent quantum threat is to public-key cryptography**, particularly digital signatures and key exchange. For cryptocurrencies, digital signatures are the linchpin of ownership: if they break, the entire system breaks. An attacker with a quantum computer could *retroactively* steal funds from any address that has ever exposed a public key – which in Bitcoin is the majority of addresses that have been used (as users often re-use addresses or use them once and leave change). As researchers noted, because the blockchain is an immutable public record, an attacker can store all historical public keys and attack them at leisure once capable. The motto *"store now, decrypt (or in this case, sign) later"* is very relevant. Even if quantum computers are a decade away, any transaction made today could be vulnerable in the future if its security is solely based on currently used primitives. This especially affects any data meant to remain confidential or unforgeable for many years (consider someone who locks bitcoins in a time-locked contract to be spent in 20 years – will ECDSA still be safe then?).

Timeline and Realistic Risk Assessment

How soon could this all happen? There are varying opinions. Some experts believe **quantum computing is still 20+ years away** from threatening public-key crypto, citing the enormous engineering challenges in scaling qubits with low error rates. Others are more optimistic, pointing to recent breakthroughs and heavy investments by governments and tech giants, suggesting a 10-year horizon for quantum disruption is not implausible. A 2021 Global Risk Institute report gave a 50% chance that RSA-2048 could be broken by 2031. In our references, Martinier's 2019 thesis concluded that the threat "*does not appear to be imminent*" but strongly advised starting upgrades "*as soon as possible to prevent 'store now, decrypt later' attacks*". This captures the consensus: **act now, even if the threat is not immediate**. The cost of proactive migration is far lower than the catastrophic cost if we wait too long. This is why NIST and NSA have set 2030 as a target for transitioning to quantum-safe algorithms.

One should also consider **which actors** could pose the threat. It is likely that the first quantum computers capable of breaking crypto will be in the hands of nation-states or large corporations, not individual hackers. This raises the scenario of a state-sponsored attack on a cryptocurrency network – an adversary with a quantum computer could, for example, *steal high-value targets' coins or even attempt to forge blocks/transactions*. The latter, however, has an important nuance: block creation in Bitcoin relies on proof-of-work, not a signature. So a quantum attacker can't *forge* a new valid transaction without the private key; they can steal keys, but they cannot fake a transaction from an address whose key they don't know (they'd first have to derive it via Shor's algorithm). In Ethereum's current Proof-of-Stake, validators' identities are tied to public keys – a quantum attacker could impersonate a validator by forging signatures if they got the keys, potentially wreaking havoc on consensus. In short, a quantum attacker would initially behave like a super-powered thief or impersonator rather than a network-level forger of all transactions.

Assessing Impact

If no countermeasures are taken, the **impact of a quantum attack on a major cryptocurrency would be devastating**. Public trust would likely collapse if high-profile thefts occurred with no recourse. The financial damage could be enormous – hundreds of billions of dollars of value could be compromised. Some authors warn this could trigger a broader "*global financial crisis*" if, say, Bitcoin were integrated deeply into the financial system by that time and suddenly collapsed due to a break in cryptography. Even the *threat* of such an event might cause loss of confidence well in advance.

It's important to note that **there are mitigations** that can reduce risk *before* full quantum readiness. For example, users can already practice **quantum hygiene**: not reusing addresses and using hash-based pay-to-script-hash (P2SH) or other constructions so that funds are always in "hash form" on-chain (only hashed pubkey, not revealed pubkey, until the moment of spending). In Bitcoin, if one never reuses an address and moves all funds to new addresses after each use, an attacker has a very narrow time window to attack (they would have to derive the private key in between the transaction being broadcast and it getting confirmed and spent by the next). This window might be as short as 10 minutes or less – which even a quantum computer would struggle with for strong curves. Thus, **address reuse** will be even more dangerous in a quantum scenario, whereas diligent users who treat addresses as single-use could mitigate some risk. Still, this is not a complete solution, and it certainly doesn't help with long-term hodling (sooner or later you'll need to spend or move, exposing a key).

In conclusion, the quantum threat to cryptocurrencies is multifaceted: it can break signatures (ownership), potentially give mining

advantages, and reduce hashing security. The most urgent concern is signature breakage via Shor's algorithm. The timeline is uncertain but likely within the next 10–20 years. Accordingly, the community and protocol developers must **prepare well in advance**. In the next section, we turn to exactly that: the development of **post-quantum cryptography and how blockchain systems can transition to it**, ensuring that cryptocurrencies remain secure against quantum-capable adversaries.

Post-Quantum Cryptography and Blockchain Security

Given the threats outlined, a natural question arises: **How can we secure cryptocurrencies against quantum attacks?** The answer lies in **Post-Quantum Cryptography (PQC)** – cryptographic algorithms believed to be secure against both classical and quantum attackers. In this section, we discuss the main families of PQC algorithms, ongoing standardization efforts, and how such algorithms can be integrated into blockchain systems. We also consider the practical challenges of transitioning large, decentralized networks to new cryptographic standards, and how to do so without undermining usability or security during the process.

Post-Quantum Cryptographic Algorithms

Over the past two decades, researchers have developed numerous candidate algorithms for the post-quantum era. These typically rely on mathematical problems that, as far as we know, are not efficiently solvable by quantum (or classical) computers. Major families include:

- **Lattice-Based Cryptography:** Based on problems like Learning With Errors (LWE) or Short Integer Solutions (SIS) in high-dimensional lattices. Lattice problems are currently the frontrunners in PQC. They can be used to construct encryption (e.g., **CRYSTALS-Kyber** algorithm) and signature schemes (e.g., **CRYSTALS-Dilithium, Falcon**). These have relatively good performance and key sizes. For example, Dilithium and Falcon (both lattice-based) were selected by NIST in 2022-2023 as the primary post-quantum signature algorithms for standardization. Lattice-based schemes are attractive for blockchains: they can be made fairly efficient and some, like Falcon-512, have compact signatures (~666 bytes) and public keys (~897 bytes), which is larger than ECDSA's ~32-byte keys and ~64-byte signatures, but still practical.
- **Hash-Based Signatures:** These use only hash functions (like SHA-2/SHA-3) and are well-understood. **Merkle signature schemes** (e.g., XMSS, LMS) are stateless or stateful hash-based signatures that are already standards for certain applications. They produce quite large signatures (often several kilobytes) and each key can only sign a limited number of messages (in stateful schemes). However, they are *very secure* assuming the hash is secure, even against quantum attack (Grover's algorithm only weakens the hash a bit). Some cryptocurrencies (like the Quantum Resistant Ledger, QRL) have adopted XMSS for their addresses. Hash-based sigs could be viable in blockchain for UTXO-based systems if managed carefully (maybe one-time addresses, etc.), but their size and statefulness issues are drawbacks for widespread use (imagine having to track which parts of your Merkle tree are used).
- **Code-Based Cryptography:** Based on error-correcting codes (e.g., McEliece cryptosystem). These have very large public keys (hundreds of kilobytes) which make them less

appealing for something like blockchain use (imagine every address having a 100KB pubkey!). Nevertheless, code-based signatures like the recent **GUI-enabled HQC** or others exist, but none have been as efficient as lattice or hash-based alternatives for signatures. McEliece encryption was a finalist in NIST's process (still under consideration) and is extremely secure (it's withstood attacks for 40+ years), but the key size issue is huge.

- **Multivariate Quadratic (MQ) Signatures:** Based on solving systems of multivariate quadratic equations (an NP-hard problem). Examples are UOV or Rainbow (though Rainbow got broken during NIST competition). Some MQ systems had efficient signatures, but many have been cryptanalyzed. At the moment, none of the leading NIST candidates are MQ-based due to security concerns.
- **Supersingular Isogeny Crypto:** Based on mathematical structures in elliptic curves different from discrete logs (supersingular isogenies). SIKE was a promising candidate for key exchange with very small keys, but it was broken by an algebraic attack in 2022 (a classical attack). So isogeny-based crypto is mostly off the table for now, though some variants might survive.

NIST's PQC competition (2016–2022) evaluated dozens of candidates and by 2022 announced the first standardization choices: for encryption/KEM (Key Encapsulation Mechanism), **CRYSTALS-Kyber** was chosen; for signatures, **CRYSTALS-Dilithium**, **Falcon**, and **SPHINCS+** (SPHINCS+ is a stateless hash-based signature) were selected or kept as alternates. Falcon, notably, was chosen for its small signature size (a benefit in blockchain applications) despite being a bit more complex to implement. The process also recognized that diversity is good – having multiple algorithm types reduces the risk if one gets broken. It's recommended that blockchain systems adopt at least one primary PQC algorithm and perhaps keep an alternative as backup (just as many systems today support multiple signature schemes like both ECDSA and EdDSA).

Integration into Cryptocurrencies

Upgrading a cryptocurrency to use PQC involves changing its signature scheme (and possibly its hash if one worries about Grover, though that is less urgent). This is not trivial: it usually requires a **hard fork** or at least a **soft fork** if done cleverly (e.g., introducing new address types for PQC and encouraging users to move to them). Bitcoin's scripting system could potentially allow new signature opcodes (for Dilithium, etc.) to be added via a soft fork, as Stewart *et al.* described. Their approach was to create a new output type that uses a commit-reveal scheme to lock funds until the user demonstrates control of a PQC key. Another approach is simply defining new address versions (say, a Bech32m encoding that indicates a post-quantum public key of a certain type). For example, one could introduce a **Bitcoin-NG (Next-Gen)** address that contains, say, a Dilithium public key hashed. Users could then send funds to either legacy addresses (ECDSA) or new PQC addresses. Initially, both would be allowed. Over time, as users transition, miners could start discouraging or even charging higher fees for legacy addresses, and eventually, the community could phase them out. This gradual approach ensures no immediate cut-off or risk of stranding funds.

Ethereum and other smart contract platforms face an even more complex task: their entire smart contract logic might depend on specific cryptography (e.g., many Ethereum contracts use EC Recover for verifying signatures). To move to PQC, the virtual machines might need new precompiled contracts or opcode changes to verify PQ signatures. Allende *et al.* (2023) tackled this by implementing three

different on-chain verification methods for Falcon signatures: (i) in smart contract code (which is slow), (ii) via a modified EVM opcode (faster, but requires protocol change), and (iii) via a precompiled contract (a hybrid approach). They found that a precompiled contract or opcode gave much better performance for verification than doing it at the Solidity level. Their successful implementation on a test network demonstrated that an existing blockchain can be upgraded to support PQC at all layers – network, consensus, and application – with manageable overhead. Figures in their study showed some increase in memory and CPU usage for nodes handling PQ signatures, but not orders of magnitude more. This is encouraging: it indicates we can have quantum security with only moderate impacts on scalability.

Performance and Size Considerations

One of the main concerns bringing PQC to crypto platforms is the larger **size of keys and signatures**, which can bloat the blockchain and increase bandwidth/storage requirements. For instance, a Bitcoin transaction today with one input and two outputs might be ~250 bytes (most of that is the ECDSA signature). Replacing that with a Dilithium signature (~2–3KB) would balloon the transaction to >2KB, roughly an order of magnitude increase. This could impact block sizes and throughput. One mitigation could be to increase block size or use more efficient signature aggregation. Some PQC algorithms allow **batch verification** or aggregation (though not as straightforwardly as EC signatures can be aggregated in schemes like Schnorr). Research is ongoing on how to compress PQ signatures or use multi-signature techniques to reduce overhead. Until those are mature, blockchains might need to accept some increase in average transaction size. It's a worthwhile trade-off for security, but it has to be planned (perhaps combined with other scaling solutions like off-chain channels or rollups to offset the size increase).

Another angle is hybrid approaches

during a transition period, a cryptocurrency might use a **dual-signature scheme** – requiring both a classical and a PQ signature on each transaction. This belt-and-suspenders approach ensures security against classical attacks (since classical algorithms are very well-tested) while also protecting against quantum. However, it doubles the signature data to include both, which is quite costly in size. Gao *et al.*'s notion of a double-signature in their PQB scheme is somewhat related: they had a “first-signature” and “last-signature” to reduce correlation between message and signature (a technique to improve security), but one could imagine a first-signature using ECDSA and a second using PQC. Allende *et al.* essentially did a form of this by adding a Falcon signature while keeping the original ECDSA for Ethereum transactions. They noted that this framework can be applied to other chains, and it likely will be a strategy in practice: initially, use hybrid signatures to build confidence in PQC, then phase out the classical one when ready.

Post-Quantum Privacy

It's worth mentioning that privacy mechanisms also need to be revisited in a quantum context. For example, **Mixing and CoinJoin** techniques could be vulnerable if quantum computers can easily solve the combinatorial problem of demixing (though that's not a straightforward application of quantum algorithms). More clearly, **zero-knowledge proofs (ZKPs)** used in protocols like Zcash's shielded transactions rely on certain hardness assumptions (some use elliptic curves pairings, etc.) which might be at risk. However, many ZKPs use hashes and symmetric primitives which are mostly safe (just slightly weakened by Grover). As PQC matures, we will see **post-quantum zero-knowledge proofs** emerge (some already exist using

lattices), ensuring that privacy-preserving transactions remain secure under quantum attack.

Another interesting development is the idea of using **quantum communication (QKD)** to secure parts of the blockchain (like communication between nodes). NSA and others have pointed out that QKD is expensive and not easily scalable, so the prevailing wisdom is to stick to software-based PQC solutions. But one could envision high-value blockchain infrastructure (say inter-exchange communication or signing of multi-sig wallets by institutions) leveraging QKD channels for an extra layer of security. Nonetheless, this is likely outside the realm of end-user cryptocurrency operations, so we focus on software PQC.

Transition Strategy and Governance

The transition to PQC in a decentralized network is as much a social/governance challenge as a technical one. Convincing millions of users to upgrade their software and potentially change their wallet addresses is non-trivial. Stewart *et al.*'s "slow defense" approach cleverly allows users to secure their funds even if they act after a quantum attack becomes evident (by then assuming a soft fork is in place). But ideally, we want to migrate **before** an attack. One strategy is to announce well in advance that by year X, a network upgrade will disable classical-crypto addresses. This requires coordination across the community and miners/validators. It might be contentious if some argue the quantum threat is not imminent and that the costs (in performance) are not justified yet. This could potentially create splits (imagine a scenario where one group wants a hard fork to PQC, others resist – resulting in two coins, one quantum-safe, one not). Hopefully, awareness of the existential nature of the threat will lead to rough consensus in favor of upgrading, especially as we get closer to the time. Community education (making it clear that "if we don't upgrade, all our coins could be stolen at some point") will be key.

It's also possible some newer blockchain networks will be **PQC from inception** – this is already happening with projects like QRL (using XMSS) and certain experimental chains. If any of those gain traction, it puts competitive pressure on incumbents to follow suit. There could also be regulatory push: governments or industries might declare that after a certain date, any cryptographic system must be quantum-resistant (financial institutions are already being advised to inventory and plan upgrades for all their crypto systems). This could trickle down to public blockchains indirectly, e.g., an ETF or bank custody of crypto might demand the asset is quantum-safe.

In summary, **post-quantum cryptography provides the tools to secure blockchains against quantum computers**, but integrating those tools requires careful consideration of performance, size, and the transition process. Research and trials (like the works of Gao et al. and Allende et al.) have shown it's feasible to create a quantum-resistant blockchain or upgrade an existing one. The main algorithms have been or are being standardized by NIST. The next steps are engineering: implementing these in all major cryptographic libraries, then in blockchain clients, and finally activating them via network upgrades. Fortunately, this upgrade can be staged gradually – for instance, introduce PQC as an option first (perhaps offering bounties for people to try breaking it to build confidence), then move towards making it the default. All of this must be done while **keeping the user experience as seamless as possible**. Ideally, users should not even notice that their wallet has moved from ECDSA to, say, Dilithium under the hood, aside from maybe longer keys displayed or slightly larger QR codes. If the user experience and security can align – making the more secure choice also the easy choice – then the transition stands a much better chance of success.

Discussion

Bringing together the threads of the previous sections, it's evident that ensuring the long-term security and privacy of cryptocurrencies is a multidisciplinary challenge. We must address human factors alongside cryptographic innovation. In this discussion, we reflect on a few overarching themes and open issues: (1) the interplay between usability and advanced security measures, (2) the urgency of proactive upgrades in the face of uncertain quantum timelines, and (3) potential impacts on the broader financial system and what that implies for policy and coordination.

Usability vs. Security – Can We Have Both?

A recurring theme is the classic security dilemma: more security often means more friction for the user. However, this need not always be the case. A major goal in usable security research is to **design systems where security is intuitive or even invisible to the user**. Cryptocurrencies started with a very tech-savvy user base willing to handle complexity, but as they reach a mainstream audience, the designs must adapt. One promising approach is **smart defaults and automation**. For example, wallets can automatically create multiple key backups (encrypted under the user's password) and deposit them in the user's cloud storage or email (with user consent), providing recovery options without the user manually writing down seeds. This introduces some centralization risk (cloud compromise), but with strong encryption, the trade-off may be worthwhile for many users. Another approach is **multi-factor authentication for blockchain transactions** – something users are already familiar with in other contexts. Projects have explored using mobile devices + hardware wallet in tandem (two devices must approve a transaction), which could stop many attacks (the chance an attacker has malware on both your phone and your hardware wallet is low). If integrated well, it might not diminish UX drastically – e.g., you confirm on your phone which you likely have with you, in addition to pressing a button on your hardware wallet.

From the privacy angle, **better user education** is needed as well – but education alone isn't enough; tools must be improved. As one participant in a study said, *"Bitcoin is inherently complex... the idea of public key cryptography should be taught in school"*. While integrating basic crypto literacy into education is a great idea, we should also make it so that users don't need a PhD in cryptography to use Bitcoin privately. Integration of privacy features (like **CoinJoin-by-default** where every transaction a wallet makes is a CoinJoin) could normalize privacy. Of course, that has implications on scalability and a need for coordination among users, but initiatives like Wasabi and Samourai Wallet's default mixing, or Monero's enforced ring signatures, show that protocol-level privacy can be made transparent to the user. There's a line of academic work on **human-centered cryptographic protocols** that could be applied more in blockchain development. For instance, secure messaging apps like Signal have made cryptography invisible (it just works in the background). Can we do the same for secure wallet operations? Perhaps, if wallets take on more responsibility (and thus need to be extremely trustworthy) – like automatically detecting address reuse, warning the user or preventing it; automatically spreading funds among multiple addresses to reduce risk; or using **account abstraction** (as in Ethereum's roadmap) to allow social recovery and other user-friendly features at the protocol level.

Quantum Readiness – When to Pull the Trigger?

A critical discussion point is timing. Upgrading systems for PQC too early might impose unnecessary costs if quantum computers are still far off; upgrading too late could be catastrophic. The window for safe procrastination is closing. NIST and government agencies are basically saying “start now” (as per NSA’s guidance, start transitioning by 2025, finish by 2030). Cryptocurrencies, being decentralized, don’t have a top-down mandate to do so, which means leadership must come from influential developers, community figures, and maybe economic majority (large holders, exchanges) to push it. We expect that as soon as NIST finalizes standards (which happened in 2024 for first algorithms) , major software libraries (OpenSSL, etc.) will incorporate them. Not long after, we should see Bitcoin Improvement Proposals (BIPs) or Ethereum Improvement Proposals discussing PQC. Indeed, discussions have already occurred on forums like the Bitcoin development mailing list about quantum resistance, with some proposing concrete steps (one reference shows Tim Ruffing’s 2018 suggestion on bitcoin-dev regarding such transition).

One open question: **What if a sudden quantum breakthrough happens?** If, say, in 2028 a lab announces a functioning 5000-qubit error-corrected computer (enough to break 256-bit EC), would the networks be able to respond quickly? A fast-track emergency hard fork might be needed. But coordinating that under panic could be messy (and it might already be too late for some addresses). This scenario underscores why having a *slow, steady* transition beforehand is preferable. Stewart et al.’s defense mechanism was slow (deliberately, to be sure it works even post-break) , but it assumes you’ve prepared the protocol changes in advance. It would be wise for blockchain communities to implement the necessary code for PQC support **well before** it’s mandatory, essentially having it in their back pocket.

Security Incidents and Public Perception:

Every major crypto hack or theft that happens (due to usability lapses or otherwise) has a detrimental effect on public perception and trust. If users keep hearing about people losing fortunes due to lost passwords or hacks, they might shy away from participating. Conversely, if quantum fears hit headlines (“Quantum computer cracks Bitcoin wallet”), it could either cause panic or spur action. Interestingly, recent surveys of businesses indicate extreme concern about quantum threats to encryption . We might see a time when holding crypto on non-quantum-safe addresses is seen as an irresponsible risk (perhaps insurance companies or custodians will mandate PQC usage). This external pressure could accelerate adoption of post-quantum addresses.

Policy and Collaboration:

Unlike early days, cryptocurrencies don’t exist in isolation; companies like exchanges, wallet providers, and even governments (for CBDCs or just holding crypto) have a stake. An industry consortium or working group on **Crypto Quantum Readiness** could be beneficial to coordinate standards (e.g., agree on which PQC algorithms to use to avoid fragmentation). We wouldn’t want Bitcoin picking one algorithm and Ethereum another if they could converge (though having two different ones also adds diversity). Standards bodies like IEEE or IETF might also step in to guide how to encode PQ keys in common formats, etc. The **open-source community** will be crucial – implementations of PQC must be vetted and optimized. It’s worth noting that introducing new cryptography can introduce new **bugs** (there have been instances of implementation flaws in new schemes). So the transition period must include a lot of testing and maybe bug bounty programs to ensure the new PQC features don’t inadvertently weaken security through side channels or mistakes.

Financial System Impact:

If major cryptocurrencies successfully transition to quantum-safe algorithms and enhance their privacy and usability, they will likely cement their role in the future financial system. However, failure to do so could marginalize them. For example, if Bitcoin were slow to upgrade and a quantum scare happened, perhaps people would flock to an alternative crypto that is seen as safer. This could shuffle the hierarchy of cryptocurrencies. On the other hand, if done right, the public might barely notice beyond some news that “Bitcoin has upgraded its cryptography to fend off quantum hackers.” That narrative could actually increase confidence and highlight the technology’s adaptability.

Remaining Technical Challenges

There are still technical hurdles being worked on:

- Efficient **multisignature or threshold schemes** in the post-quantum world (essential for things like multi-sig wallets, consensus signatures in some protocols, etc.). Some lattice schemes allow threshold signing, but it’s more complex.
- **Quantum-resistant light clients:** SPV (simplified payment verification) clients rely on cryptographic hashes (Merkle proofs) which are fine, but they also sometimes rely on bloom filters or other stuff that might be affected in a quantum context. Generally fine, but needs a look.
- **Interoperability:** during the transition, how do quantum-safe addresses interact with legacy ones? If someone sends from a PQ address to a classical address, the transaction itself might have both signature types. Ensuring seamless interoperability so that, for instance, users of old wallets can still receive funds (maybe wrapped in a way) from new wallets is tricky. Possibly everyone might be forced to upgrade – or intermediate solutions like “*two-layer addresses*” that include both an old and new public key hash to be spendable by either.

Human Factors in Transition:

One thing to consider is user *migration*. Say a new wallet version generates a Dilithium-based address. Users need to move their coins from old addresses to new ones. Some will procrastinate (just like some people still had bitcoins in old vulnerable P2PK addresses for years). How to incentivize or enforce this? Wallet software could periodically warn users: “You have funds in a legacy address that may become unsafe in the future. Click here to migrate them to a secure address.” Exchanges could also play a role: they could automatically credit users’ accounts and reissue funds to new addresses (basically sweeping customer deposits periodically). Perhaps the mining community might at some point refuse to include transactions from very old script types (though that would be a hard stance and unlikely without consensus).

Quantum as Catalyst for Improvement:

Interestingly, the impending quantum threat can be viewed as an *opportunity* to fix some longstanding issues. It provides a clear motivation to overhaul parts of the system that are otherwise left as is.

In doing so, developers might take the chance to also incorporate other improvements (like addressing some scalability or privacy issues). For example, if Bitcoin requires a hard fork to introduce PQC addresses, that might open a window to bundle other upgrades that have been desired but not urgent. This requires careful management to avoid overreach, but it's a thought.

Beyond Cryptography – Quantum Threats to Proof-of-Work:

We touched on Grover's effect on PoW. If that becomes serious, one mitigation is to move away from pure hash-based PoW. Bitcoin probably won't, but other chains might accelerate plans for Proof-of-Stake or more quantum-resistant PoW (some research on quantum-resistant PoW suggests using structures that are NP-hard problems not sped up by quantum). Alternatively, simply doubling hash size (Bitcoin could hard fork to SHA-512 perhaps) would counter Grover by making the effective security 256-bit again. But that requires caution (and hardware changes for miners).

In any case, **planning for resilience** is key. The community should consider worst-case scenarios and build systems that fail gracefully. For instance, what if a quantum attacker starts draining addresses one by one? Having a monitoring system to detect likely quantum attacks (e.g., if someone suddenly cracks an address known to be secure under classical assumptions) could alert the network. Perhaps nodes could automatically flag and delay transactions that look suspiciously like they could be quantum-generated, though distinguishing that is hard.

Conclusion

Cryptocurrencies stand at a crossroads of innovation and challenges. On one hand, they promise a decentralized, open financial system; on the other, they currently demand users to shoulder significant security burdens and face an approaching storm in the form of quantum computing. In this paper, we have argued that **enhancing usable security and privacy** is essential for cryptocurrencies to achieve widespread, safe adoption, and that this effort must go hand-in-hand with preparing for the **quantum age**.

Our analysis has highlighted that many of the security failures in today's cryptocurrency ecosystem are not due to flaws in cryptography, but due to usability issues – from lost keys to phishing scams facilitated by confusing interfaces. Addressing these requires a human-centric approach: designing wallets and protocols that minimize the scope for user error and make secure practices the path of least resistance. Encouraging trends include the development of protocols that embed security at the UI level (like user-friendly address verification and social recovery mechanisms) and community emphasis on education. Yet, more work between the cryptocurrency community and HCI (Human-Computer Interaction) experts is needed to create truly **user-resilient systems**. As one study put it, current systems “*are not resilient to human errors*” – future ones should strive to be.

Simultaneously, the impending advent of quantum computing adds a ticking clock to these improvements. We have surveyed the state-of-the-art in quantum attacks and defenses, showing that while quantum computers pose a serious threat to current cryptographic protocols, viable post-quantum alternatives exist and are maturing rapidly. The path forward for cryptocurrencies involves adopting these quantum-resistant algorithms *before* large quantum computers arrive. The research and trials done so far – from lattice-based signature schemes to full frameworks for quantum-resistant blockchains – give confidence that this migration is technically achievable. The remaining

challenges are in optimization, standardization, and deployment at scale, all of which are actively being worked on as part of global PQC initiatives.

Key recommendations emerging from this work are:

1. **Begin Transition to Post-Quantum Cryptography Early:** Cryptocurrency developers and communities should not wait until quantum computers are breaking encryption. The next few years offer a critical window to test and roll out post-quantum signature schemes (e.g., through soft forks introducing new address types). By 2030, as per industry and government timelines, all major public blockchains should aim to have quantum-resistant options in use, if not have fully migrated. Starting early allows for ironing out implementation bugs and ensuring a smooth transition for users.
1. **Adopt a Hybrid Security Approach During Migration:** For a period, using hybrid schemes (classical + PQC signatures) or gradual phasing as described can hedge risks. While this increases transaction sizes temporarily, it provides assurance against both classical and quantum attacks. Users should be encouraged via wallets to move funds to new dual-key addresses in upcoming releases.
1. **Invest in Wallet UX and User Education:** Innovations like integrated key backup solutions, more intuitive identity/address systems (to reduce mistakes), and privacy-by-default should be prioritized in wallet development. At the same time, clear messaging about security best practices (like not reusing addresses, using hardware wallets, etc.) and the reasons behind upcoming changes (like PQC) will help users acclimate and trust the system. The community should treat usability as a first-class feature, as important as throughput or latency.
2. **Collaborate on Standards:** Stakeholders across different cryptocurrency projects should collaborate (perhaps via open standards forums) on selecting and standardizing PQC algorithms for blockchain use. This includes agreeing on formats for PQC public keys and signatures in transactions, so that inter-chain solutions or future cross-chain operations are easier. A concerted effort can also better liaise with NIST and academic experts to ensure the chosen algorithms are robust and well-implemented.
1. **Monitor Quantum Progress and Adjust Plans:** Keeping an eye on developments in quantum computing is crucial. If breakthroughs accelerate, contingency plans (like fast deprecation of vulnerable addresses) should be ready. Conversely, if quantum progress stalls, the pace of transition can be adjusted – though not halted, since the threat would still loom. Essentially, maintain an agile posture informed by the latest science.
1. **Enhance Privacy Mechanisms:** In tandem with security, privacy needs to be maintained in a quantum world. Post-quantum privacy techniques (like PQ zero-knowledge proofs) should be integrated once available, and interim

measures like larger mixing sets or improved coinjoin implementations used. Privacy and fungibility are important for cryptocurrencies' value proposition and user safety (from other threats like surveillance), and they should not be forgotten in the rush to quantum-proof.

We have also drawn attention to the fact that many security solutions introduce complexity that can backfire if not carefully implemented. As noted in one review, solutions that **add complexity and administrative overhead can inadvertently create new usability challenges**. This caution is especially relevant when adding new cryptographic systems. Thus, part of our recommendation is thorough testing and user feedback loops – e.g., when a post-quantum wallet is released, have usability studies to see how non-expert users cope with any differences. Perhaps the ideal outcome is that the user hardly notices the shift, aside from maybe software update notes.

In concluding, we reiterate that cryptocurrency systems are unique in that they place control (and responsibility) directly in the hands of users. This empowers individuals but also exposes the system to the full range of human fallibility. Strengthening usable security is therefore not only about protecting individuals but about safeguarding the system's collective security (since one weak link can have system-wide effects, as in 51% attacks aided by social engineering, or large-scale theft undermining confidence). The coming quantum era raises the stakes – it is a fast-approaching external threat that demands unity and action. However, it also provides an impetus – a chance to upgrade and refine cryptocurrency networks to be more secure, private, and robust than ever before. With careful planning, collaboration, and emphasis on the human element, cryptocurrency can not only survive the quantum age, but thrive in it as a cornerstone of a secure, user-empowered digital economy.

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